

SWYNEX

SECURITY ASSESSMENT REPORT

Defensive Security Assessment, Web Security & Network Traffic Analysis

Scope	Authorized training/lab environments documented in the supplied reports
Primary lab	Kali Linux / VMware / Metasploitable 2
Web security	Apache HTTPD, Apache Tomcat 5.5, DVWA
Network analysis	Wireshark 4.6.6 — network_security_lab.pcapng
Date	07 October 2026

Purpose

This report consolidates the supplied Security Fundamentals Assessment, Web Security Review, Wireshark Network Security Analysis, and the provided screenshots into one defensive Task 4 submission report.

1. Executive Summary

The supplied assessment material documents defensive security testing in intentionally vulnerable training environments. The combined assessment identifies exposed and legacy services, vulnerable software, anonymous file-sharing access, legacy web components, a reflected cross-site scripting (XSS) condition, and multiple exposed web/management surfaces. The supplied reports state that the testing was limited to authorized local lab environments and that no exploit was executed in the Security Fundamentals assessment.

The Wireshark exercise complements the host and application findings with network-level observations. The capture contains ARP, DNS, ICMP, TCP and TLS 1.3 traffic. The evidence demonstrates protocol behavior and provides a basis for defensive monitoring.

The highest-priority remediation areas are obsolete/vulnerable software, anonymous access, exposed management interfaces, the confirmed reflected-XSS condition, unnecessary attack surface, and replacement of insecure Telnet administration.

Assessment at a glance

Area	Key result	Highest documented severity
Core services	Telnet, vsftpd 2.3.4, Apache/PHP and anonymous SMB	Critical
Web security	Legacy Apache/Tomcat, reflected XSS and exposed management surfaces	High
Network analysis	DNS, ICMP, TCP and TLS 1.3 observed	Observation-based

2. Scope, Environment and Methodology

Authorized scope. The Security Fundamentals report describes an isolated VMware training lab with Kali Linux as the assessment machine and Metasploitable 2 as the target. The documented assessment IPs are 192.168.174.129 and 192.168.174.128. The Web Security Review separately documents an authorized local lab target at 192.168.126.128 (Metasploitable2/DVWA).

Methodology. Nmap service/version detection was used for host and service discovery. SearchSploit references were reviewed for the detected vsftpd version. HTTP response headers were reviewed for Apache/PHP versions. SMB shares were enumerated without credentials. The DVWA reflected-XSS condition was validated with a harmless proof-of-concept. Wireshark filters were used for DNS, ICMP, TCP and TLS analysis.

Tools. Nmap, SearchSploit, Netcat, curl, smbclient, browser validation, Kali Linux, VMware Workstation and Wireshark 4.6.6.

The supplied Security Fundamentals report states that no exploitation was performed; this report therefore emphasizes identification, evidence, risk and remediation.

3. Core Security Findings

These findings come from the supplied Security Fundamentals Assessment. Severity labels are preserved from the source report.

F-01 — Telnet Service Exposed	Medium
Evidence / impact TCP/23 was identified as an open Linux telnet service.	Remediation Disable Telnet where not required; use SSH and restrict administrative access.
F-02 — Vulnerable vsftpd 2.3.4 Service	Critical
Evidence / impact TCP/21 was identified as vsftpd 2.3.4. SearchSploit returned references for the documented backdoor command-execution issue. No exploit was executed.	Remediation Remove/upgrade vsftpd; restrict FTP exposure; prefer secure alternatives such as SFTP.
F-03 — Outdated Apache and PHP Web Stack	High
Evidence / impact TCP/80 was reported as Apache 2.2.8 and HTTP headers disclosed PHP 5.2.4.	Remediation Upgrade Apache/PHP, apply updates, remove unnecessary modules and minimize version disclosure.
F-04 — Anonymous SMB Access to tmp Share	High
Evidence / impact Samba was identified on TCP/139 and TCP/445. Anonymous login succeeded and the tmp share could be listed.	Remediation Disable guest/anonymous access unless required; require authentication and least privilege.

4. Web Security Assessment

This section is explicitly based on the supplied SWYNEX-Web-Security-Review. The assessment target is the authorized local training VM 192.168.126.128 (Metasploitable2 / DVWA). The review used Nmap, browser validation and screenshots.

4.1 Web security findings

WEB-01 — Legacy Apache HTTP Server 2.2.8	High
Evidence / impact Nmap identified TCP/80 open and reported Apache httpd 2.2.8 ((Ubuntu) DAV/2). Legacy web-server software increases exposure to known security weaknesses.	Remediation Upgrade to a supported Apache release; remove unnecessary modules/features; apply security updates.
WEB-02 — Legacy Apache Tomcat 5.5 on TCP/8180	High
Evidence / impact Nmap reported Apache Tomcat/Coyote on TCP/8180. The browser displayed the Tomcat 5.5 default page with Status, Administration and Manager links.	Remediation Upgrade Tomcat; restrict management interfaces; remove default/example applications; enforce strong authentication.
WEB-03 — Reflected Cross-Site Scripting (XSS) in DVWA	High
Evidence / impact A harmless alert('XSS') payload was supplied to DVWA and the browser displayed the resulting alert, confirming script execution in the application page context.	Remediation Use context-aware output encoding, safe DOM APIs, appropriate validation, restrictive CSP and secure cookie attributes.
WEB-04 — Multiple web applications and management surfaces exposed	Medium
Evidence / impact The HTTP landing page exposed TWiki, phpMyAdmin, Mutillidae, DVWA and WebDAV; the Tomcat page exposed administration and manager entry points.	Remediation Disable/remove unnecessary applications and expose management interfaces only through a protected administrative network or VPN.

4.2 Web security evidence

The supplied Web Security Review contains evidence for Nmap service/version discovery, the Metasploitable2 web landing page, the Tomcat 5.5 default page, and the DVWA reflected-XSS proof of concept. The extracted evidence images below are included to make the final report self-contained.

```
thaila@kali:~$ nmap -sV -p 80,443,8080,8180 192.168.126.128
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 02:34 -0400
Nmap scan report for 192.168.126.128
Host is up (0.00081s latency).

PORT      STATE SERVICE      VERSION
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
443/tcp   closed https
8080/tcp   closed http-proxy
8180/tcp   open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:61:54:86 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.34 seconds
```

Figure 6 — Nmap service/version discovery.

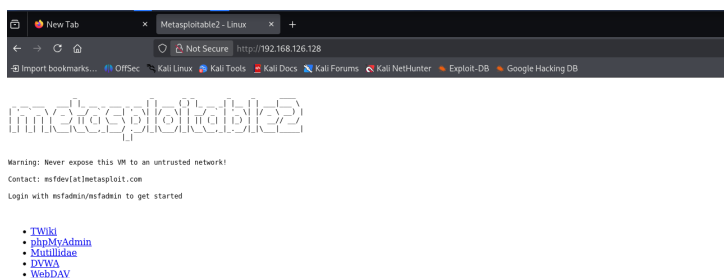


Figure 7 — Metasploitable2 web landing page.

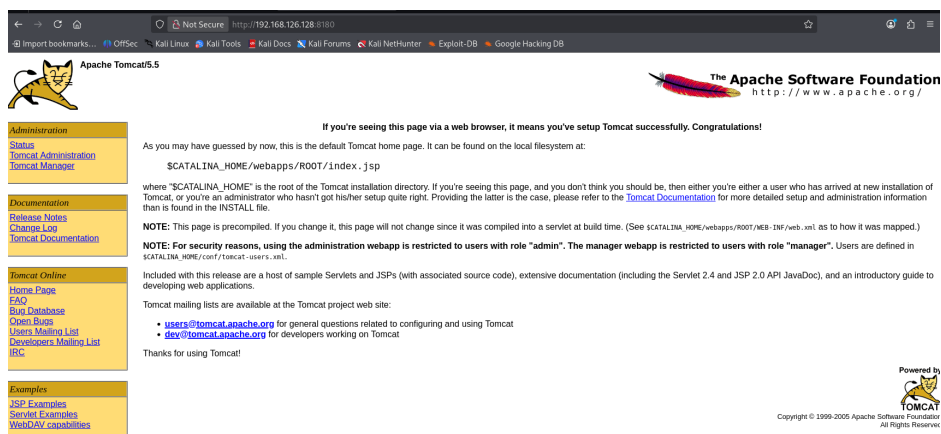


Figure 8 — Apache Tomcat 5.5 default page / exposed management links.

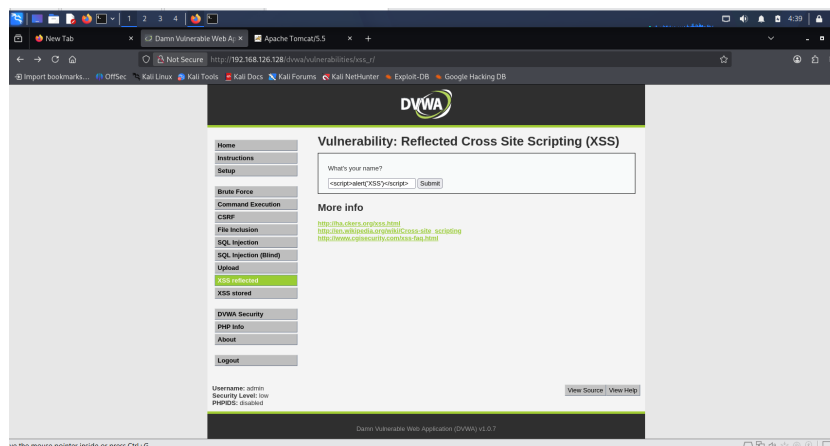


Figure 9 — DVWA reflected XSS page with harmless payload.

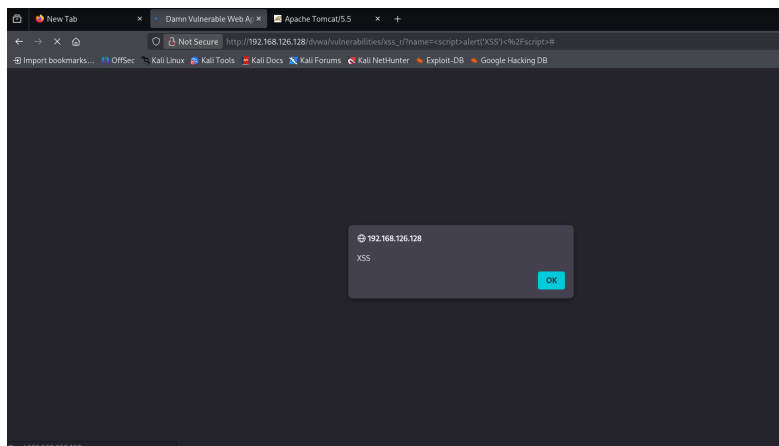


Figure 10 — DVWA reflected XSS proof-of-concept execution.

5. Wireshark Network Traffic Analysis

The supplied Wireshark report documents a defensive analysis using Kali Linux, Wireshark 4.6.6, interface eth0 and network_security_lab.pcapng. The following screenshots are the provided packet-capture evidence.

5.1 Overall capture

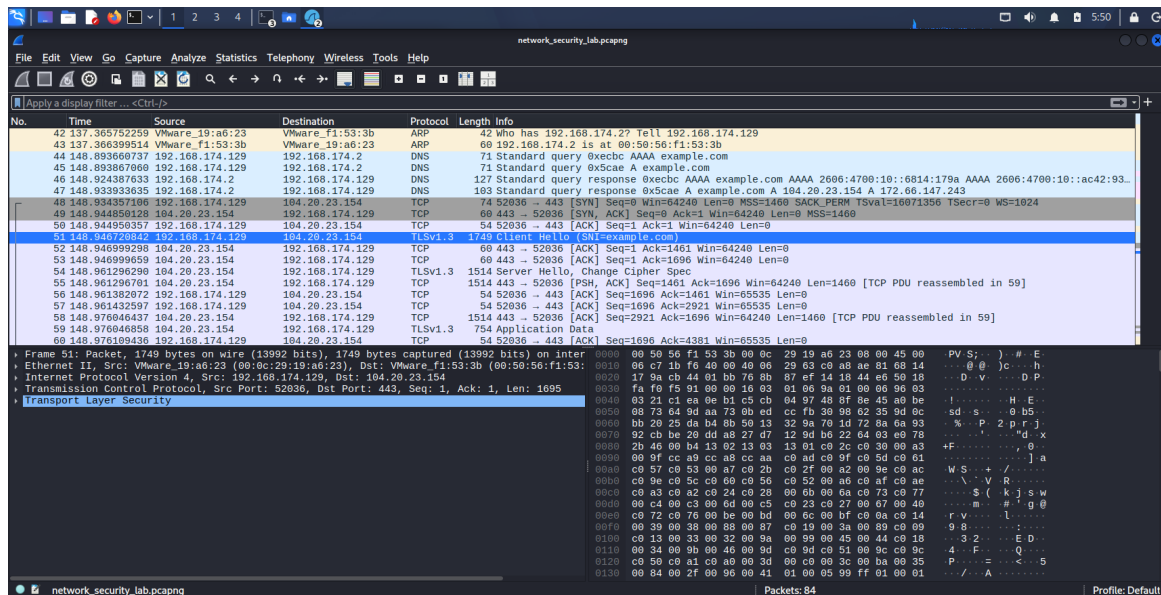


Figure 11 — Overall Wireshark capture.

5.2 DNS analysis

DNS queries and responses involving example.com were observed and are consistent with normal name resolution in the supplied analysis.

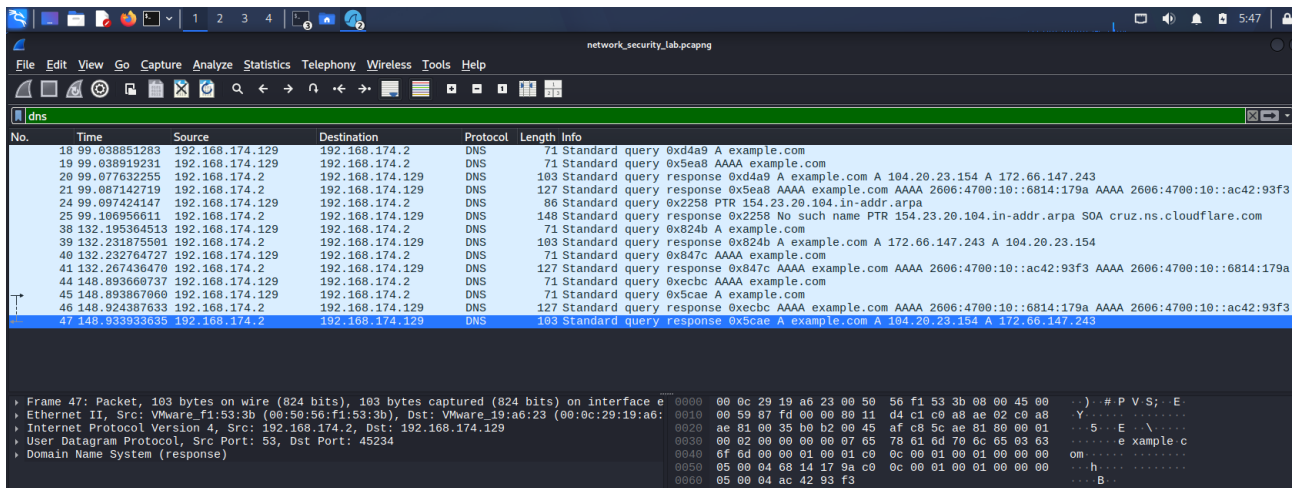


Figure 12 — DNS query/response traffic.

5.3 ICMP analysis

ICMP Echo Requests and Replies were observed between 192.168.174.129 and 104.20.23.154, demonstrating successful connectivity.

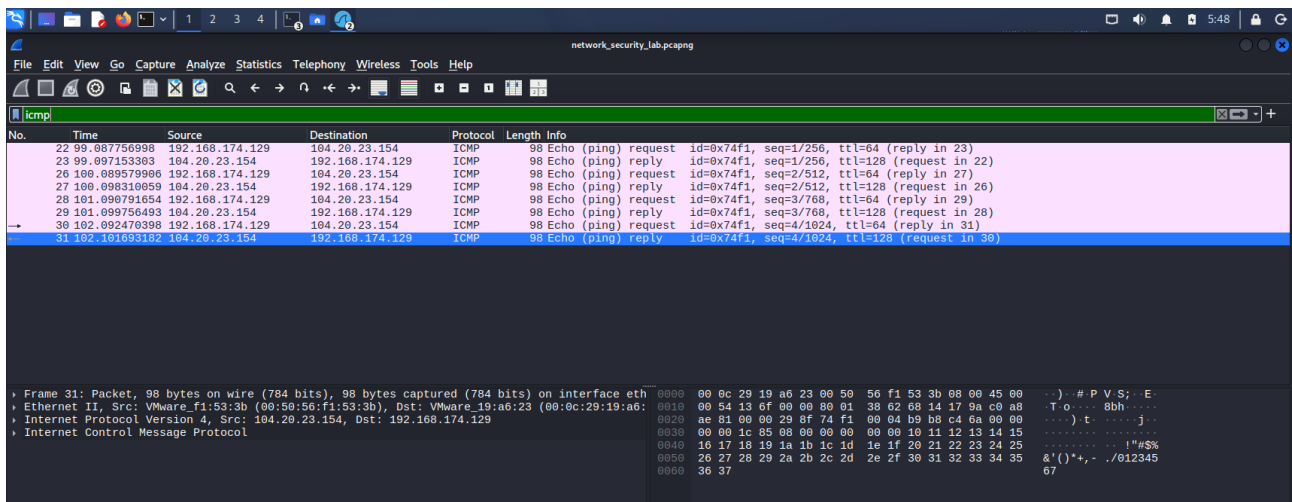


Figure 13 — ICMP Echo Request/Reply traffic.

5.4 TCP analysis

TCP traffic using destination port 443 was observed. The connection begins with the standard SYN, SYN-ACK and ACK three-way handshake followed by data exchange.

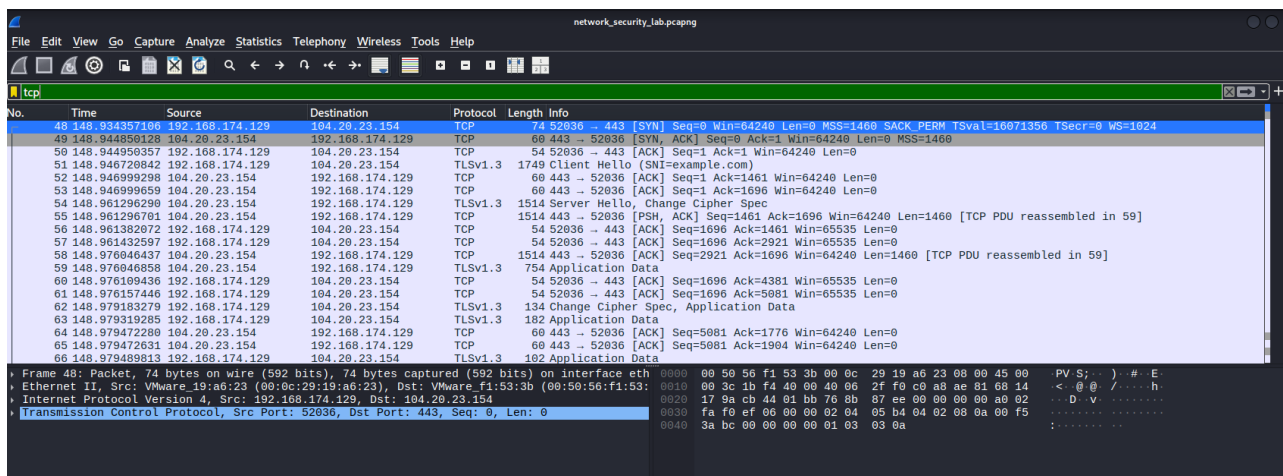


Figure 14 — TCP connection establishment and port 443 traffic.

5.5 TLS 1.3 analysis

The capture contains TLS 1.3 Client Hello, Server Hello, Change Cipher Spec and encrypted Application Data. The Client Hello identifies example.com through SNI.

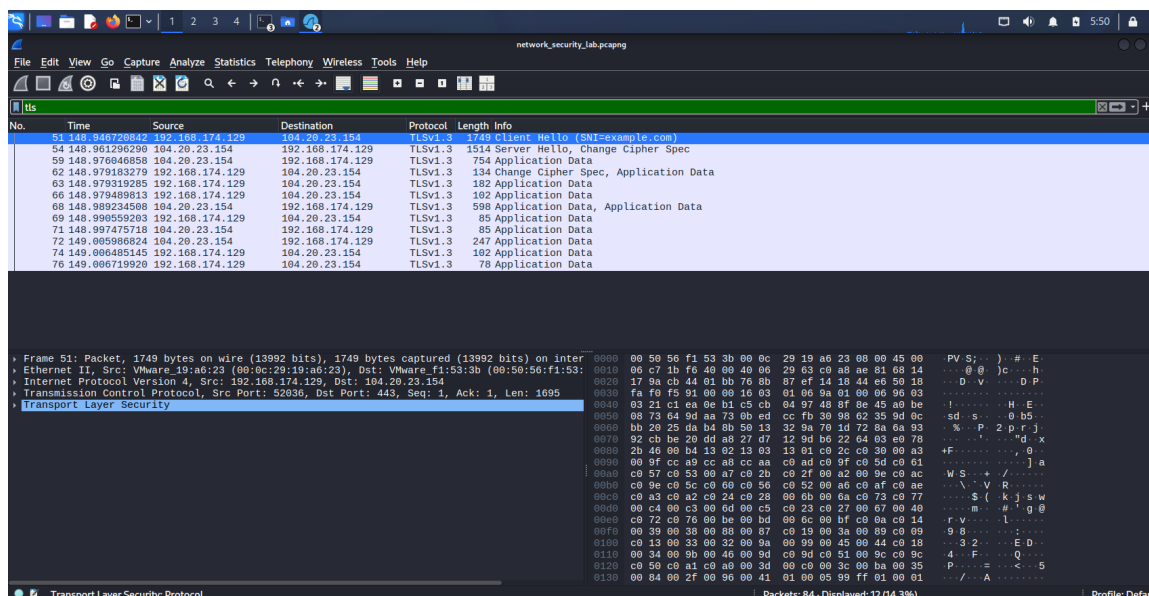


Figure 15 — TLS 1.3 handshake and encrypted application traffic.

Network-security interpretation

The Wireshark capture does not by itself establish a vulnerability. It provides protocol and metadata evidence useful for defensive monitoring. Unexpected destinations, unusual DNS queries, high-volume ICMP, repeated failed connections, unusual ports or anomalous encrypted traffic would warrant investigation.

6. Consolidated Risk Summary

Priority	Finding	Severity
1	F-02 — vsftpd 2.3.4	Critical
2	F-04 — Anonymous SMB	High
3	F-03 — Outdated Apache/PHP	High
4	WEB-01 — Legacy Apache 2.2.8	High
5	WEB-02 — Legacy Tomcat 5.5	High
6	WEB-03 — Reflected XSS	High
7	F-01 — Telnet	Medium
8	WEB-04 — Multiple surfaces	Medium

6.1 Recommended remediation plan

- Immediately isolate the training VM from untrusted networks.
- Upgrade or retire Apache 2.2.8, Tomcat 5.5 and other obsolete components.
- Remove or upgrade vsftpd 2.3.4 and restrict FTP exposure.
- Disable anonymous SMB access and enforce least privilege.
- Restrict Tomcat Administration/Manager endpoints to trusted administrators.
- Remove default/example applications and unnecessary services.
- Fix reflected XSS using context-aware output encoding and safe DOM practices; add restrictive CSP and secure cookie attributes.
- Disable Telnet and use SSH for secure remote administration.
- Repeat service discovery and application security testing after remediation.

7. Conclusion

The consolidated assessment demonstrates defensive security analysis across host services, web applications and network traffic. The strongest documented risks are the vulnerable vsftpd 2.3.4 service, anonymous SMB access, obsolete Apache/PHP/Tomcat components, reflected XSS and unnecessary web/management exposure.

The Wireshark evidence adds network-level visibility into DNS, ICMP, TCP and TLS 1.3 behavior. The recommended response is to reduce attack surface, maintain supported software, restrict administrative and anonymous access, correct application-layer XSS, and monitor network activity for anomalies.

All findings and evidence in this report are based on the supplied authorized training-lab materials. No additional external targets or unsupported vulnerabilities were introduced during preparation.

Prepared for SWYNEX Internship — Task 4